

Appendix of EMA

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A THEORETICAL GUARANTEES OF EMA

In this section, we show that EMA preserves the asymptotic complexity of the base index in terms of both construction time and index size. We further analyze the source of false positives introduced by Markers and provide theoretical bounds on the false positive rate.

A.1 Complexity

LEMMA A.1 (COST OF MEncode AND EDGE EXISTANCE). *In Algorithm 3, MEncode takes $O(m)$ time. Testing whether $e_{(n,v)}$ previously exists takes $O(M)$ time, since it scans at most M previously stored neighbors.*

THEOREM A.2. *The time complexity of Algorithm 3 is*

$$O(efc \cdot M \cdot (M + m)).$$

PROOF. The outer loop iterates over candidates $v \in Cand$, so it executes at most $|Cand| \leq efc$ times.

For a fixed candidate v , the inner loop scans $u \in Nbrs$. The neighbor list is bounded by the maximal size M , hence $|Nbrs| \leq M$ and the inner loop runs at most M times.

Consider one inner-loop iteration for a pair (u, v) . Besides a constant-time distance comparison, when u dominates v the algorithm (i) tests whether $e_{(n,v)}$ previously exists, which costs $O(M)$, and then (ii) calls either Marker merge or MEncode, which costs $O(m)$ by Lemma A.1. Therefore, each inner-loop iteration costs $O(M + m)$ in the worst case.

Thus, for each candidate v , the inner loop costs

$$O(M \cdot (M + m)).$$

The remaining work per candidate (insertion into $Nbrs$ and one additional MEncode) is at most $O(M + m)$ and is dominated by the inner-loop bound. Summing over all $|Cand| \leq efc$ candidates gives

$$O(efc \cdot M \cdot (M + m)).$$

□

The time complexity of HNSW index construction is empirically close to $O(n \log n)$ and is highly correlated with the parameters M and efc , which can be expressed as $O(M \cdot efc \cdot n \log n)$. In our

method, Marker propagation introduces additional operations, resulting in a complexity of $O(M \cdot (M + m) \cdot efc \cdot n \log n)$. However, since both M and m are constants that are significantly smaller than n , the asymptotic time complexity remains unchanged.

THEOREM A.3. *The space complexity of EMA is*

$$O(nM \cdot sm),$$

PROOF. The additional storage introduced by EMA consists of the Marker stored per neighbor (edge), which is only used in the bottom layer.

In the bottom layer, each of the n nodes stores at most M neighbors, resulting in at most nM edges. Each edge stores a Marker of size sm bits, so the total Marker storage is

$$(nM) \cdot sm \text{ bits.}$$

The top layer stores $O(\log n)$ neighbors per node and does not store Markers and therefore introduce no additional space over the original HNSW index.

Combining both parts yields a total space complexity of

$$O(nM \cdot sm + M \log n),$$

which simplifies to $O(Mn \cdot sm)$. □

A.2 Theoretical Analysis of Marker False Positives

Definition A.4 (Expected Case-(1) False Positive Rate). Consider a directed edge $e_{(u,v)}$ whose Marker is constructed by aggregating the attribute bitmaps of the dominating node v and the set of nodes dominated by v with respect to u , denoted by $D(e)$; hence $v \notin D(e)$.

Let R be a query predicate with selectivity sel , i.e., for a randomly drawn node p ,

$$\Pr[p \models R] = sel,$$

and assume that predicate satisfaction events are independent and identically distributed across nodes.

A Case-(1) false positive occurs on edge e if the Marker is activated due to at least one dominated node satisfying the predicate, while the target node itself does not satisfy the predicate. Formally,

$$FP_{case1}(e) \iff (v.A \not\models R) \wedge (\exists p \in D(e) : p \models R).$$

THEOREM A.5 (EXPECTED CASE-(1) FALSE POSITIVE RATE). *Let $|D(e)|$ denote the dominate size of edge e , and let*

$$\mu = \mathbb{E}[|D(e)|]$$

be the average dominate size over all edges.

Under the assumptions of Definition A.4, the expected false positive rate induced by Case (1) is given by

$$FPR_{case1} = (1 - sel) \cdot \left(1 - \mathbb{E}[(1 - sel)^{|D(e)|}]\right).$$

Moreover, the false positive rate admits the following upper bound that depends only on the average dominate size μ :

$$0 \leq FPR_{case1} \leq (1 - sel) \cdot (1 - (1 - sel)^\mu).$$

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In particular, under the common average-size approximation $|D(e)| \approx \mu$, we obtain

$$\text{FPR}_{\text{case1}} \approx (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^\mu).$$

PROOF. By definition, a Case-(1) false positive occurs when the Marker of an edge $e = (u \rightarrow v)$ is activated due to at least one dominated node satisfying the query predicate, while the target node v itself does not satisfy the predicate. Formally, the event is

$$\text{FP}_{\text{case1}}(e) \iff (v.A \not\models R) \wedge (\exists p \in D(e) : p \models R).$$

Conditioned on a fixed dominate size $|D(e)| = k$, independence implies

$$\Pr[v.A \not\models R] = 1 - \text{sel},$$

and

$$\Pr(\exists p \in D(e) : p \models R) = 1 - \Pr(\forall p \in D(e) : p \not\models R) = 1 - (1 - \text{sel})^k.$$

Since the two events are independent, we obtain

$$\Pr(\text{FP}_{\text{case1}} \mid |D(e)| = k) = (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^k).$$

Taking expectation over the random variable $|D(e)|$ yields

$$\Pr(\text{FP}_{\text{case1}}) = (1 - \text{sel}) \cdot \left(1 - \mathbb{E}[(1 - \text{sel})^{|D(e)|}]\right).$$

To derive a bound that depends only on the average dominate size $\mu = \mathbb{E}[|D(e)|]$, observe that the function $f(k) = (1 - \text{sel})^k$ is convex for $k \geq 0$. By Jensen's inequality,

$$\mathbb{E}[(1 - \text{sel})^{|D(e)|}] \geq (1 - \text{sel})^{\mathbb{E}[|D(e)|]} = (1 - \text{sel})^\mu.$$

Substituting this inequality gives

$$\text{FPR}_{\text{case1}} \leq (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^\mu).$$

Non-negativity of probabilities establishes the lower bound. $\square$

Next, we derive a bound to control Codebook-induced false positives from case (2) and determine an appropriate Codebook size s for a given selectivity range.

Let $R = \bigwedge_{j=1}^m R_j$ be a conjunctive predicate on m attributes, and let

$$\mathcal{R} = \{v \in \mathcal{D} \mid R(v) = \text{true}\}$$

denote the set of records satisfying R .

Let $\tilde{\mathcal{R}}$ denote the set of records accepted by Marker checking, i.e.,

$$\tilde{\mathcal{R}} = \{v \in \mathcal{D} \mid \exists_u \text{MMatch}(e_{(u,v)}).\text{Marker} = \text{true}\},$$

which are the records that are reachable through at least one Marker-compatible edge during joint search.

Define the Case-(2) false positive rate as

$$\text{FPR}_{\text{case2}}(R) = \frac{\Pr[v \in \tilde{\mathcal{R}} \setminus \mathcal{R}]}{\Pr[v \in \tilde{\mathcal{R}}]}.$$

THEOREM A.6 (BOUNDING CASE-(2) FALSE POSITIVES). Assume that:

- (i) For each attribute $A^{(j)}$, the Codebook partitions its domain into s disjoint regions, each with probability mass at most $1/s$;
- (ii) For each attribute $A^{(j)}$, the predicate R_j partially intersects at most b_j Codebook regions.

If the joint predicate selectivity satisfies $\Pr[v \in \mathcal{R}] \geq \text{sel}$, then

$$\text{FPR}_{\text{case2}}(R) \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

THEOREM A.7 (CODEBOOK SIZE FOR CONTROLLING CASE-(2) FALSE POSITIVES). If the joint predicate satisfies $\Pr[x \in \mathcal{R}] \geq \text{sel}$, then

$$\text{FPR}_{\text{case2}}(\mathcal{R}) \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

To guarantee $\text{FPR}_{\text{case2}}(\mathcal{R}) \leq \text{FP}$, it suffices that

$$s \geq \frac{(1 - \text{FP})}{\text{FP} \cdot \text{sel}} \cdot \sum_{j=1}^m b_j.$$

PROOF. For each attribute dimension $A^{(j)}$, $j = 1, \dots, m$, the Marker-induced region $\tilde{R}_j$ is a superset of $R_j(A^{(j)})$. The only contribution to $\tilde{R}_j \setminus R_j(A^{(j)})$ comes from regions that intersect $R_j(A^{(j)})$ but are not fully contained in it. By assumption, there are at most b_j such regions, and each contributes at most $\rho \leq 1/s$ probability mass. Therefore,

$$\Pr[a_j \in \tilde{R}_j \setminus R_j(A^{(j)})] \leq \frac{b_j}{s}.$$

Let E_j denote the event $a_j \in \tilde{R}_j \setminus R_j(A^{(j)})$. Since

$$\tilde{\mathcal{R}} \setminus \mathcal{R} \subseteq \bigcup_{j=1}^m E_j,$$

the union bound gives

$$\Pr[x \in \tilde{\mathcal{R}} \setminus \mathcal{R}] \leq \sum_{j=1}^m \Pr[E_j] \leq \sum_{j=1}^m \frac{b_j}{s}.$$

Moreover, since $\tilde{\mathcal{R}}$ is a superset of $\mathcal{R}$,

$$\Pr[x \in \tilde{\mathcal{R}}] \geq \Pr[x \in \mathcal{R}] \geq \text{sel}.$$

Combining the above inequalities yields

$$\text{FPR}_{\text{case2}}(\mathcal{R}) = \frac{\Pr[x \in \tilde{\mathcal{R}} \setminus \mathcal{R}]}{\Pr[x \in \tilde{\mathcal{R}}]} \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

Requiring the right-hand side to be at most FP gives

$$(1 - \text{FP}) \sum_{j=1}^m \frac{b_j}{s} \leq \text{FP} \cdot \text{sel},$$

which is equivalent to the stated sufficient condition on s . $\square$

Example. Consider a conjunctive predicate $\mathcal{R} = \bigcap_{j=1}^m R_j(A^{(j)})$ on m attributes, and assume each $R_j(A^{(j)})$ is a 1D range predicate under a contiguous Codebook partition, so that $b_j \leq 2$. Then $\sum_{j=1}^m b_j \leq 2m$, and Theorem A.7 gives the sufficient condition

$$s \geq \frac{(1 - \text{FP})}{\text{FP} \cdot \text{sel}} \cdot 2m.$$

For instance, with $m = 4$, $\text{sel} = 50\%$, and $\text{FP} = 50\%$, it suffices to use $s \geq 16$ bits per attribute. If $\text{sel} = 10\%$ and $\text{FP} = 50\%$, the bound becomes $s \geq 80$.

These theorems establish the computational and theoretical foundations of EMA. We analyze the index-time and space complexity of EMA, and provide a principled characterization of false positives

arising from dominance-based aggregation (Case 1) and codebook granularity (Case 2). Together, these results clarify when marker-based pruning is effective and show how appropriate codebook sizing can bound granularity-induced false positives while preserving efficient graph traversal.

B DYNAMIC SUPPORT

Table 1 reports the per-step cost of EMA dynamic operations on SIFT10M. Deletions are highly efficient: mark-delete incurs negligible overhead (1.3 s per 1M), and attribute-only updates are similarly lightweight (2.5 s), as they only modify local metadata without affecting graph connectivity. Insertions incur moderate cost, comparable to standard ANN index construction, and remain practical under parallel execution.

C COMPLEX PREDICATE COMBINATION

We evaluate EMA on complex Boolean predicates that combine numerical ranges and categorical constraints. Specifically, we use the following predicate:

$$p = (\text{num} \in [a_1, b_1] \wedge \text{cate} \supseteq L_1) \vee (\text{num} \in [a_1, b_2] \wedge \text{cate} \supseteq L_2),$$

which captures a representative class of composed predicates with mixed attribute types and AND/OR operators. We vary the selectivity from 1% to 10% while fixing recall at 95% (recall@10).

Figure 1 reports the results. Under controlled selectivity, EMA maintains stable QPS across different predicate compositions, demonstrating its robustness in handling arbitrary Boolean combinations over numerical and categorical attributes.

D FAIR COMPARISON WITH HNSWLIB AND NAVIX

Since HNSWLib natively supports filtering, we include it as a baseline to demonstrate the effectiveness of EMA. In addition, NaviX is implemented on top of Faiss, which is generally less competitive than HNSWLib. To ensure a fair comparison, we evaluate (i) naive filtering in HNSWLib and (ii) NaviX implemented on top of HNSWLib.

Figure 6

Figure 2a shows that EMA consistently outperforms HNSWLib. Moreover, NaviX does not achieve better performance than the HNSWLib baseline.

Figure 2b reports recall-QPS trade-offs at 10% selectivity. We implement a prototype of NaviX on HNSWLib. Although HNSWLib alone achieves competitive performance, NaviX is sensitive to the underlying index behavior. Compared to Faiss, HNSWLib explores more candidates at the bottom layer; the additional two-hop expansion further amplifies random memory accesses, leading to degraded performance. A detailed analysis will be included in the revision.

E PARAMETER SENSITIVITY ANALYSIS

E.1 Effect of Graph Degree M

We evaluate the impact of the maximum graph degree $M \in \{16, 32, 40, 64\}$. All other parameters are fixed ($\text{ft_bits} = 128$, $\text{ft_routing_min_deg} = 8$, $e_f c = 300$), using single-threaded execution over 1,000 queries.

Table 2 reports QPS at 95% recall, where bold entries indicate the best M for each selectivity.

Overall, the optimal choice of M depends on selectivity. At low selectivity (1%–3%), smaller M performs best, as it reduces neighbor comparison cost and enables faster transition to edge recovery. Increasing M brings no benefit in this regime due to limited candidate expansion. At moderate selectivity (10%–40%), $M = 32$ achieves the highest QPS, balancing search quality and traversal overhead. At high selectivity (80%–100%), $M = 40$ becomes optimal, as slightly larger neighborhoods improve connectivity without incurring excessive scanning.

Notably, overly large M (e.g., 64) consistently underperforms due to increased traversal and unnecessary neighbor evaluation. These results suggest that moderate values of M provide the best trade-off between exploration cost and graph connectivity across selectivity levels.

E.2 Effect of FT Width (ft_bits)

We evaluate the impact of ft_bits on performance. Figure 3 reports QPS at 95% recall across selectivity.

$\text{ft_bits}=128$ consistently achieves the best performance, improving over 32 bits by 2%–25%. Increasing from 32 to 128 bits reduces false positives and improves routing accuracy, while further increasing to 256 bits yields no benefit due to the enlarged memory footprint, which increases cache access cost and dominates query latency. $\text{ft_bits}=64$ provides a reasonable trade-off but remains slightly inferior to 128 bits. Thus, we use $\text{ft_bits}=128$ by default.

E.3 Effect of Edge Recovery Threshold (min_deg)

We evaluate the effect of the minimum degree in edge recovery (min_deg). Figure 4 reports QPS at 95% recall under varying selectivity.

Performance exhibits a clear peak at moderate min_deg . Under $M=40$ (level-0 degree ≈ 80), the optimal range lies between 15 and 20, achieving the highest QPS across selectivities. Increasing min_deg beyond this range leads to a sharp degradation (30%–55%), as the search progressively approaches post-filtering behavior.

When min_deg becomes large (e.g., ≥ 40), the effective neighborhood is saturated, and traversal degenerates to visiting nearly all neighbors, effectively discarding FT-based routing. This regime is consistently slower (up to 2.9 $\times$) than the optimal setting, highlighting that FT routing is effective only when min_deg is properly constrained.

Overall, these results reveal a narrow operating region ($\text{min_deg} \in [8, 20]$) where EMA significantly outperforms post-filtering, confirming that controlled edge recovery is critical to preserving routing efficiency.

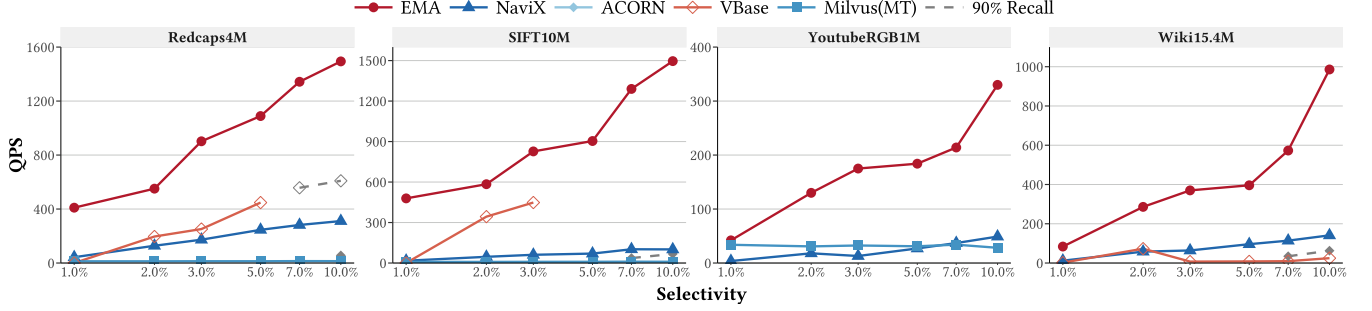
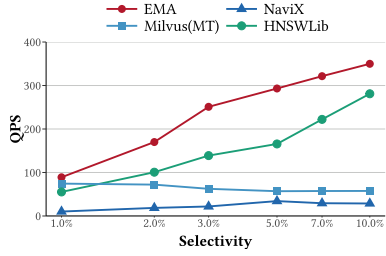
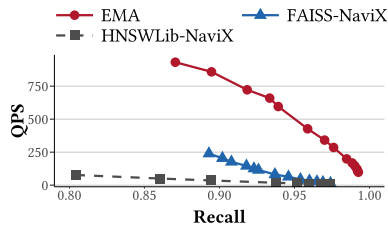
F OTHER STUDIES

F.1 False positive rate of the Marker.

The effectiveness of EMA depends on the accuracy of the Marker, as a lower false positive rate (FPR) reduces unnecessary candidate exploration during joint filtering. Table 3 reports the FPR on RedCaps1M across selectivity levels ranging from 1% to 60%.

Table 1: Per-step average cost of EMA dynamic operations on SIFT10M.

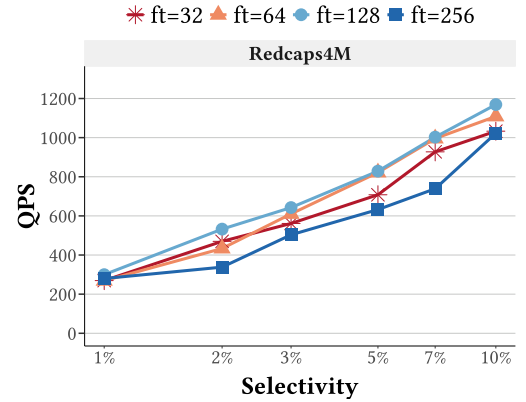
Step	Avg. cost	Mode	Threads
Insert (add_items, per 1 M)	90.8 s	parallel	32
Mark-delete (per 1 M)	1.3 s	serial	1
Attr-only update (update_attr, per 1 M)	2.5 s	serial	1
Update vec+attr (per 1 M)	140.5 s	parallel	32
Patch (batched_patch_deletes, per call)	45.2 s	parallel	32
Reconstruct (5 M survivors → fresh index)	371.1 s	parallel	32

**Figure 1: QPS for composed multi-predicate queries at 1%–10% selectivity with 95% recall@10.****(a) QPS vs. selectivity on YoutubeRGB1M at 95% recall.****(b) QPS vs. recall on YoutubeRGB1M at 10% selectivity.****Figure 2: Experiments across different baselines.**

The reported false positive rates are measured with edge recovery disabled and a fixed search parameter (efs), so as to isolate the accuracy of the Marker itself. Across this broad range of selectivities, the FPR consistently remains below 10% for both range+label and range-only queries. As selectivity increases, the FPR grows moderately, which is expected as filtering becomes less restrictive.

Table 2: HashANN QPS at recall=0.95 on Redcaps_4M across selectivity and graph degree M .

M	1%	3%	10%	40%	80%	100%
16	375	635	931	2200	2958	3765
32	308	591	1112	2606	3097	4904
40	330	576	1073	2575	3436	5438
64	286	533	1056	2364	3399	4870
best M	16	16	32	32	40	40

**Figure 3: Impact of ft_bits on QPS at 95% recall across different selectivity levels.**

Overall, these results indicate that the Marker provides effective pre-checking with a low and stable false positive rate, supporting efficient query processing across different selectivity regimes.

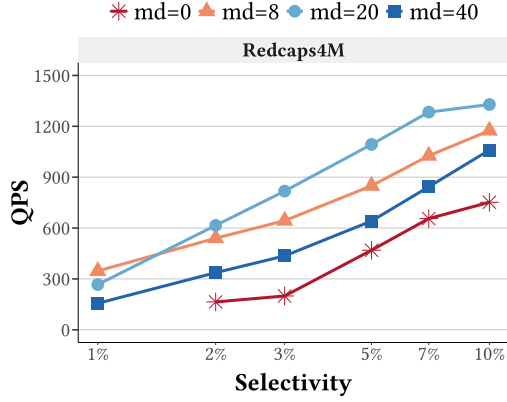


Figure 4: Impact of min_deg on QPS at 95% recall across different min_deg levels.

Table 3: FPR on Redcaps1M.

Selectivity	FPR	
	Range+Label	Range
1%	3.84%	2.99%
5%	4.80%	3.67%
10%	5.41%	4.27%
20%	6.20%	5.18%
60%	7.97%	7.71%

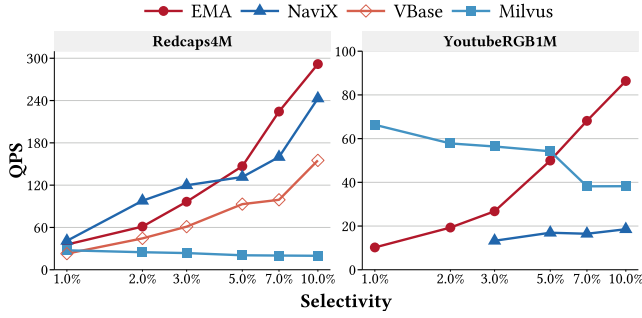


Figure 5: Label-range query at 95% recall@100.

F.2 Performance at 95% Recall@100.

Figure 5 reports the QPS under the $R@100$ setting. While EMA exhibits slightly lower throughput in some cases, it is important to note that EMA consistently achieves a higher recall level (approximately 99%) compared to the 95% recall target used by other methods.

This difference arises from the more conservative convergence criteria adopted by EMA, which prioritize retrieval accuracy and result stability. As a result, the reported throughput reflects a stricter recall requirement, highlighting a trade-off between efficiency and accuracy under the $Rrecall@100$ evaluation.

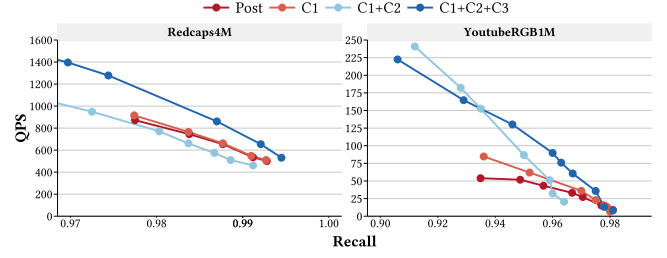


Figure 6: Ablation study on Redcaps and Youtube-RGB.

G ABLATION STUDY.

Building upon the classic post-filtering approach on HNSW, we decompose our method into three key components: (1) diversity-aware pruning (C1), (2) joint filtering with Markers (C2), and (3) neighbor recovery under low selectivity (C3).

Figure 6 presents an ablation study on the YouTube-RGB dataset under 1% selectivity and the RedCaps dataset under 10% selectivity with both range and label constraints. Compared with the original HNSW post-filtering baseline, each proposed component consistently improves retrieval performance, and their combination yields the best overall results. In particular, introducing the Marker slightly reduces QPS at high recall targets, but yields a noticeable performance improvement at lower recall levels (around 92%). This improvement indicates that the Marker effectively prunes unnecessary traversals while preserving sufficient candidates, providing a strong foundation for further performance gains when combined with the edge recovery strategy.